

Retrieval-Augmented Generation for Knowledge-Intensive NL

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Abstract

We present a comprehensive study of retrieval-augmented generation (RAG) systems. Our work examines the trade-offs between dense and sparse retrieval methods, and demonstrates that combining them via reciprocal rank fusion significantly improves retrieval quality. We evaluate multiple reranking strategies and show their impact on downstream generation quality. Our experiments cover dense retrieval models like DPR, sparse methods like BM25, and hybrid approaches. We also introduce Self-RAG, a method that incorporates self-reflection to improve grounding.

1 Introduction

Retrieval-Augmented Generation (RAG) has emerged as a powerful paradigm for combining the parametric knowledge of large language models with non-parametric external knowledge sources. RAG models generate output by first retrieving relevant documents from a corpus and then conditioning the language model on the retrieved information.

Dense retrieval models like DPR (Dense Passage Retrieval) encode queries and passages into a shared dense vector space and retrieve nearest neighbors using maximum inner product search. BM25, on the other hand, is a sparse retrieval method based on term frequency statistics.

Hybrid retrieval combines the strengths of both dense and sparse approaches. Dense retrieval captures semantic similarity while sparse retrieval excels at exact term matching. Reciprocal Rank Fusion (RRF) is a principled method for combining these approaches into a unified candidate set.

Reranking further improves retrieval quality by using cross-encoders that jointly encode the query and each candidate passage. This approach considers fine-grained interactions between the query and each candidate but is computationally more expensive than bi-encoder approaches.

Self-RAG is a variant that incorporates self-reflection into the RAG pipeline. It generates reflection tokens that allow the model to critique its own retrieval and generation steps, leading to more grounded responses.

2 Related Work

2.1 Dense Retrieval

DPR encodes queries and passages using separate bi-encoders and retrieves the most similar passages via vector similarity. This approach captures semantic relationships but may miss exact term matches.

2.2 Sparse Retrieval

BM25 is a classical sparse retrieval method based on term frequency and inverse document frequency statistics. It excels at exact term matching but lacks semantic understanding.

2.3 Hybrid Approaches

Recent work has explored combining dense and sparse retrieval. RRF provides a principled way to fuse ranking lists from multiple retrievers without tuning combination parameters.

3 Methodology

3.1 Hybrid Retrieval Pipeline

Our hybrid retrieval pipeline first retrieves top-k candidates using both dense and sparse retrievers. We then apply reciprocal rank fusion to combine the results into a unified candidate set. A cross-encoder reranker then re-scores the candidates based on query-candidate interaction.

3.2 Reranking

The reranker uses a cross-encoder architecture that jointly encodes the query and each candidate passage. This allows the model to capture fine-grained relevance signals that are not available in bi-encoder retrieval.

References

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